File Handling using File Streams

File represents storage medium for storing data or information. Streams refer to sequence of bytes. In Files we store data i.e. text or binary data permanently and use these data to read or write in the form of input output operations by transferring bytes of data. So we use the term File Streams/File handling. We use the header file `<fstream>`

- **ofstream:** It represents output Stream and this is used for writing in files.
- **ifstream:** It represents input Stream and this is used for reading from files.
- **fstream:** It represents both output Stream and input Stream. So it can read from files and write to files.

Operations in File Handling:

- Creating a file: `open()`
- Reading data: `read()`
- Writing new data: `write()`
- Closing a file: `close()`

Creating/Opening a File

We create/open a file by specifying new path of the file and mode of operation. Operations can be reading, writing, appending and truncating. Syntax for file creation: `FilePointer.open("Path",ios::mode);`

```
//CREATE
#include<iostream>
#include<conio>
#include <fstream>
using namespace std;
int main()
{
   fstream st; // Step 1: Creating object of fstream class
   st.open("E:\MyFile.txt",ios::out);  // Step 2: Creating new file
   if(!st) // Step 3: Checking whether file exist
   {
      cout<<"File creation failed";
   }
   else
   {
      cout<<"New file created";
      st.close(); // Step 4: Closing file
   }
   getch();
   return 0;
}
```

```
//WRITE
#include <iostream>
#include<conio>
#include <fstream>

using namespace std;

int main()
{
   fstream st; // Step 1: Creating object of fstream class
   st.open("E:\myFile.txt",ios::out);  // Step 2: Creating new file
   if(!st) // Step 3: Checking whether file exist
   {
      cout<<"File creation failed";
   }
   else
   {
      cout<<"New file created";
      st<<"Hello";    // Step 4: Writing to file
      st.close(); // Step 5: Closing file
   }
   getch();
   return 0;
}
```

```
//READ

#include <iostream>
#include<conio>
#include <fstream>

using namespace std;

int main()
{
   fstream st; // step 1: Creating object of fstream class
   st.open("E:\myFile.txt",ios::in);   // Step 2: Creating new file
   if(!st) // Step 3: Checking whether file exist
   {
      cout<<"No such file";
   }
   else
   {
      char ch;
      while (!st.eof())
      {
         st >>ch;  // Step 4: Reading from file
         cout << ch;   // Message Read from file
      }
      st.close(); // Step 5: Closing file
   }
   getch();
   return 0;
}
```